

Project Title

Exploring-household-products

Project Objective:

The focus of this assignment is on data summarization, conditional logic, and text extraction

techniques, which are essential for preparing data for further analysis.

Dataset Description:

Attributes: Product ID, Product Name, Brand Name, Quantity, Category, Price

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
28-JAN-US	Laptop	Dell	1000	30	Electronics
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	Smartphone	Samsung	900	25	Electronics
22-MAY-US	Backpack	North Face	70	20	Outdoor
07-JUN-UK	Headphones	Sony	200	45	Electronics
19-JUL-UK	T-shirt	Adidas	30	5	Fashion
23-AUG-UK	Blender	Ninja	90	35	Kitchen
05-SEP-UK	Tablet	Apple	500	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
17-JUN-IN	Laptop	HP	950	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics
16-APR-ES	Headphones	Bose	250	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics
27-JAN-IT	Laptop	Asus	980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor
14-MAY-RU	Camera	Nikon	700	50	Electronics
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories

29-SEP-CA	Smartphone	Google	800	45	Electronics
03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion
11-JUL-CA	Blender	Vitamix	400	40	Kitchen
16-APR-ES	Headphones	Bose	230	20	Electronics
07-MAR-CA	Dress	Zara	60	30	Fashion
13-APR-CA	Toaster	Hamilton	40	10	Kitchen
24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics
02-DEC-CA	Jeans	Levi's	50	50	Fashion
17-JUN-IN	Laptop	HP	950	25	Electronics
09-JUL-FR	Watch	Casio	100	20	Accessories

Basic Data Exploration

1. Sum, Count, and Average

- Calculate the total price of all products in the dataset.

=SUM(D2:D35)

SUM = 10100

- Determine how many products exist in the dataset.

=COUNTA(B2:B35)

COUNT=34

- Calculate the average price of the products.

=AVERAGE(I8/34)

AVERAGE=297.1

2. Minimum and Maximum Values

- Identify the minimum price among all products.

=MIN(D2:D35)

MINIMUM= 30

Identify the maximum price among all products.

=MAX(D2:D35)

MAXIMUM=1000

3. Logical Function – IF

Create a new column named Price Range.

Column Name	Description
Ex: PRICE RANGE	HIGH PRICE OR STAND PRICE

Use the IF function to categorize products based on their price:

- If Price $\geq$ \$500 → classify as High Price
- If Price $<$ \$500 → classify as Standard Price Data cleaning & transformation:

ANSWER

=IF(D2>=500,"HIGH PRICE",IF(D2<500,"STANDART PRICE"))

4. Conditional Functions – SUMIF and COUNTIF

Perform the following conditional calculations:

- Use the SUMIF function to calculate the total price of products in the Electronics category.

=SUMIF(F2:F35,"Electronics",D2:D35)

- COUNTIF function to determine the number of products with a price less than \$100.

=COUNTIF(D2:D34, "<100")

COUNTIF= 11

5. Text Formatting – LEFT, RIGHT, MID

Create a new column named Day with the first 2 characters of each 'Product ID' using the LEFT function.

COLUMN NAME	Description
Ex: DAY	EX 28

=LEFT(A2,2)

LEFT FUNCTION

DAY=28

- Create a new column named Country Code by extracting the last 2 characters from the 'Product ID' column using the RIGHT function.

COLUMN NAME	Description
Ex: COUNTRY CODE	EX 28

=RIGHT(A2,2)

- Create a new column named Month by extracting the 4th to 6th characters from the 'Product ID' column using the MID function.

=MID(A2,3,5)

COLUMN NAME	Description
Ex: MONTH	EX JAN

Visualization and Insights Arrived with screenshot of visualization

Report/Dashboard screenshot:

Electronic price	PRICE RANGE	DAY	COUNTRY CODE
8050	HIGH PRICE	28	US
7050	STANDART PRICE	15	US
7050	STANDART PRICE	03	US
7050	HIGH PRICE	11	US
6150	STANDART PRICE	22	US
6150	STANDART PRICE	07	UK
5950	STANDART PRICE	19	UK
5950	STANDART PRICE	23	UK
5950	HIGH PRICE	05	UK
5450	STANDART PRICE	14	UK
5450	HIGH PRICE	17	IN
4500	STANDART PRICE	25	AU
4500	STANDART PRICE	08	DE
4500	STANDART PRICE	18	CA
4350	STANDART PRICE	16	ES
4100	STANDART PRICE	21	CA
4100	STANDART PRICE	20	CN
3940	HIGH PRICE	27	IT
2960	STANDART PRICE	01	UK
2960	STANDART PRICE	14	US
2960	HIGH PRICE	14	RU

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Electronic price	PRICE RANGE	DAY	COUNTRY CODE	MONTH
8050	HIGH PRICE	28	US	-JAN-
7050	STANDART PRICE	15	US	-FEB-
7050	STANDART PRICE	03	US	-MAR-
7050	HIGH PRICE	11	US	-APR-
6150	STANDART PRICE	22	US	-MAY-
6150	STANDART PRICE	07	UK	-JUN-
5950	STANDART PRICE	19	UK	-JUL-
5950	STANDART PRICE	23	UK	-AUG-
5950	HIGH PRICE	05	UK	-SEP-
5450	STANDART PRICE	14	UK	-OCT-
5450	HIGH PRICE	17	IN	-JUN-
4500	STANDART PRICE	25	AU	-NOV-
4500	STANDART PRICE	08	DE	-DEC-
4500	STANDART PRICE	18	CA	-FEB-
4350	STANDART PRICE	16	ES	-APR-
4100	STANDART PRICE	21	CA	-AUG-
4100	STANDART PRICE	20	CN	-AUG-
3940	HIGH PRICE	27	IT	-JAN-
2960	STANDART PRICE	01	UK	-MAR-
2960	STANDART PRICE	14	US	-AUG-
2960	HIGH PRICE	14	RU	-MAY-
2260	STANDART PRICE	09	CA	-JAN-
2260	STANDART PRICE	19	BR	-JUL-
2110	STANDART PRICE	21	CA	-AUG-
2110	HIGH PRICE	29	CA	-SEP-
1310	STANDART PRICE	03	CA	-JUN-
1310	STANDART PRICE	11	CA	-JUL-
1310	STANDART PRICE	16	ES	-APR-
1080	STANDART PRICE	07	CA	-MAR-
1080	STANDART PRICE	13	CA	-APR-
1080	STANDART PRICE	24	CA	-MAY-
950	STANDART PRICE	02	CA	-DEC-
950	HIGH PRICE	17	IN	-JUN-
0	STANDART PRICE	09	FR	-JUL-

Conclusion:

In conclusion, this project successfully achieved its aim of excel function. By using the excel formulas and While this project provided valuable insights into product information, valuable conclusion

